CHE 201 Fall 2019 HW 8

Sargamkumar Patel

TOTAL POINTS

90 / 100

QUESTION 1

1 1 **15 / 15**

✓ **- 0 pts** Correct

- 5 pts no unit

- 5 pts Q incorrect

- 2 pts gallon of hot water incorrect

- 15 pts no submission

- 15 pts Wrong question, problem 4.55

QUESTION 2

2 2 **7.5 / 15**

- 0 pts Correct

✓ **- 7.5 pts** **Partially wrong or incomplete solution. T2 = 759 °R and T4 = 779 °R**

- 15 pts Wrong solution. T2 = 759 °R and T4 = 779 °R

QUESTION 3

3 3 **20 / 20**

✓ **- 0 pts** Correct

- 3 pts T3 unit in K

- 10 pts Power output

- 3 pts for calculate T3, h should also included h5 and 6

- 10 pts T3 = ?

- 5 pts T3 unfinished

- 20 pts no submission

QUESTION 4

4 4 **15 / 15**

✓ **- 0 pts** Correct

- 3 pts Q is negative. Heat transfer from the system

- 2 pts Final answer should be in MW

- 5 pts Calculation error. Q = -0.21 MW

- 5 pts Delta PE and Delta KE are different than zero

- 5 pts Wrong unit conversion. Q = -0.21 MW

- 15 pts Missing or wrong answer

QUESTION 5

5 5 **15 / 15**

✓ **- 0 pts** Correct

- 5 pts Stage 1: Q = mΔu (closed system energy balance) = 4204.2 kJ

- 3 pts Stage 2: correct equation but wrong solution. Q = +8725.5 kJ. Heat transfer to the system.

- 5 pts Stage 2: Q = Δ(m.u) - heΔm = +8725.5 kJ (energy balance one-exit control volume). Heat transfer to the system.

- 15 pts Missing/wrong answer

QUESTION 6

6 6 **17.5 / 20**

- 0 pts Correct

- 1.5 pts Wrong unit conversion. m1 = 1.2 x 10^5 kg and m2 = 9.3 x 10^5 kg

- 5 pts Wrong energy balance. Should use Eq 4.28

- 5 pts W = 282 GJ

✓ **- 2.5 pts** **Work is positive (done by the compressor)**

- 20 pts Wrong answer

- 0 pts Click here to replace this description.

55) Model.

- Steady state — one inlet, one outlet
- Control volume
- $\Delta KE = \Delta PE = 0$
- $W_{cv} = 0$

mass balance — $\dot{m}_1 = \dot{m}_2 = \dot{m}$

$$0 = \dot{Q}_{cv} - \dot{W}_{cv} + \dot{m}\left((h_1 - h_2) + \left(\frac{V_1^2 - V_2^2}{2}\right) + g(z_1 - z_2)\right)$$
$$= \dot{Q}_{cv} + \dot{m}(h_1 - h_2)$$

At 90°F, $h_1 = 58.07 \ \frac{Btu}{lb}$ At 120F, $h_2 = 88 \ \frac{Btu}{lb}$

$$\dot{Q}_{cv} = \dot{Q}_{in}(SA)(0.75) \qquad 100 - 25\% = 75\%$$
$$= 200 \ \frac{Btu}{h \cdot ft^2}(24 \ ft^2)(0.75)$$
$$= 3600 \ \frac{Btu}{h}$$

$$0 = \dot{Q}_{cv} + \dot{m}(h_1 - h_2)$$
$$\therefore \dot{m} = \frac{\dot{Q}_{cv}}{(h_2 - h_1)} = \frac{3600 \ \frac{Btu}{h}}{(88 - 58.07) \ \frac{Btu}{lb}}$$
$$\therefore \dot{m} = 120.28 \ \frac{lb}{h}$$

$$\dot{m} = \frac{\dot{V}}{v} \quad \therefore \dot{V} = \dot{m}v$$
$$\therefore \dot{V}_2 = \dot{m}v_2 \qquad v_2 = 0.01621 \ \frac{ft^3}{lb}, \text{ at } 120F$$
$$\therefore \dot{V}_2 = 120.28 \ \frac{lb}{h}\left(0.01621 \ \frac{ft^3}{lb}\right)$$
$$\therefore \dot{V}_2 = 1.9497 \ \frac{ft^3}{hr}\left(\frac{1 \ gal}{0.133681 \ ft^3}\right)$$
$$\dot{V}_2 = 14.585 \ \frac{gal}{hr}.$$

1 1 **15 / 15**

- ✓ **- 0 pts** Correct
- **- 5 pts** no unit
- **- 5 pts** Q incorrect
- **- 2 pts** gallon of hot water incorrect
- **- 15 pts** no submission
- **- 15 pts** Wrong question, problem 4.55

58 Model.

- 2 inlet, 2 outlet
- $\Delta KE = \Delta PE = 0$
- Control volume
- Steady state.
- Ideal gas

$Cp_{CO_2} = 0.24004 \frac{Btu}{lb \cdot R}$

$\therefore (\dot{m} Cp dT)_{CO_2} = (\dot{m} Cp dT)_{air}$

$Cp_{air} = 0.202 \frac{Btu}{lb \cdot R}$

$\therefore \dot{m}_1 [Cp(T_1 - T_2)]_{CO_2} = \dot{m}_3 [Cp(T_3 - T_4]_{Air}.$

$\therefore 73.7 [0.24004 (560 - T_2)] = 50 [0.202 (T_2 + 20 - 1040)]$

$9906.931 - 17.691 T_2 = 10.1 T_2 - 10302$

$T_2 = 727.067 ^\circ R$ exit temperature of CO_2

$\therefore T_4 = T_2 + 20$

$= 727.067 + 20 = 747.067 ^\circ R$. exit temperature of air.

2 2 **7.5 / 15**

- **- 0 pts** Correct
- ✓ **- 7.5 pts Partially wrong or incomplete solution. T2 = 759 °R and T4 = 779 °R**
- **- 15 pts** Wrong solution. T2 = 759 °R and T4 = 779 °R

76 Model

- Steady state.
- Control volume
- $\Delta KE = \Delta PE = 0$
- $\dot{Q}_{cv} = 0$
- Ideal gas

~~At~~ $P_1 = 20\,bar, T_1 = 600^\circ C \Rightarrow h_1 = 3690.1\ \frac{kJ}{kg}$

At $P_2 = 10\,bar, T_2 = 400^\circ C, h_2 = 3263.9\ \frac{kJ}{kg}$

At $P_4 = 1\,bar, T_4 = 240^\circ C, h_4 = 2954.5\ \frac{kJ}{kg}$

Table A-22

At $T_5 = 1500K, P_5 = 1.35, h_5 = 1635.97\ \frac{kJ}{kg}$

At $T_6 = 1200K, P_6 = 1\,bar, T_6 = 1277.79\ \frac{kJ}{kg}$.

mass-balance

$\dot{m}_1 = \dot{m}_2 = \dot{m}_3 = \dot{m}_4 = \dot{m}_{steam}$

$\dot{m}_5 = \dot{m}_6 = \dot{m}_{air}$

$\therefore$ Energy-balance for turbine 1

$$0 = \dot{Q}_{cv}^{\,0} - \dot{W}_{cv} + \dot{m}_s\left((h_1 - h_2) + \left(\frac{V_1^2 - V_2^2}{2}\right)^{0} + g(z_1 - z_2)^{0}\right)$$

$$= -\dot{W}_{cv} + \dot{m}_s(h_1 - h_2)$$

$$\dot{W}_{cv} = \dot{m}_s(h_1 - h_2)$$

$$\therefore \dot{m}_s = \frac{\dot{W}_{cv}}{(h_1 - h_2)} = \frac{10000}{3690.1 - 3263.9} = 23.46\ \frac{kg}{s}$$

Energy balance for heat Exchanger

$$\dot{m}_2(h_2-h_3)+\dot{m}_5(h_5-h_6)=0$$

$$h_3=h_2+\frac{\dot{m}_5}{\dot{m}_2}(h_5-h_6)$$

$$h_3=3263.9+\frac{25}{23.46}(1635.97-1277.79)$$

$$\therefore \dot{m}_5=\left(\frac{1500}{60}\right)=25\ \mathrm{kg/s}$$

$$h_3=3645.6\ \mathrm{kJ/kg}.$$

At $P_3=10$ bar and $h_3=3645.6\ \mathrm{kJ/kg}$.

$$T_3=540+(600-540)\frac{(3645.6-3565.6)}{(3697.7-3565.6)}$$

$$=540+36.28=576.28=849.28\ \mathrm{K}$$

b) Energy balance for turbine 2.

$$0=\dot{Q}_{cv}^{\,0}-\dot{W}_{cv}+\dot{m}\left((h_3-h_4)+\left(\frac{V_3^2-V_4^2}{2}\right)^{0}+g(Z_3-Z_4)^{0}\right)$$

$$\therefore \dot{W}_{cv}=\dot{m}(h_3-h_4)$$

$$\therefore \dot{W}=23.46(3645.6-2954.5)$$

$$\dot{W}=16213\ \mathrm{KW}$$

3 3 **20 / 20**

- ✓ **- 0 pts** Correct
- **- 3 pts** T3 unit in K
- **- 10 pts** Power output
- **- 3 pts** for calculate T3, h should also included h5 and 6
- **- 10 pts** T3 = ?
- **- 5 pts** T3 unfinished
- **- 20 pts** no submission

74 Mass balance : $\dot{m}_1 = \dot{m}_2 + \dot{m}_3$

$\dot{m}_1 = 2\dot{m}_2 \; (\because \dot{m}_2 = \dot{m}_3)$

$1.5 = 2\dot{m}_2$

$\dot{m}_2 = 0.75 \text{ kg/s}$

At $P_1 = 20 \text{ bar}$, $T_1 = 360°c$, $h_1 = 3159.3 \text{ KJ/kg}$

At $P_2 = P_3 = 1 \text{ bar}$

$h_2 = h_g = 2,675.5 \text{ KJ/kg}$

$h_3 = h_f = 417.46 \text{ KJ/kg}$

Energy balance.

$$0 = \dot{Q}_{cv} - \dot{W}_{cv} + \dot{m}_1(h_1 + \tfrac{1}{2}V_1^2 + gz_1) - \dot{m}_2(h_2 + \tfrac{1}{2}V_2^2 + gz_2) - \dot{m}_3(h_3 + \tfrac{1}{2}V_3^2 + gz_3)$$

$$0 = \dot{Q}_{cv} - \dot{W}_{cv} + \dot{m}_1(h_1 + \tfrac{1}{2}V_1^2 + gz_1) - 0.5\dot{m}_1(h_2 + \tfrac{1}{2}V_2^2 + gz_2) - 0.5\dot{m}_1(h_3 + \tfrac{1}{2}V_3^2 + gz_3)$$

$$0 = \dot{Q}_{cv} - (2.2\times10^6) + 1.5((3159\times10^3) + \tfrac{1}{2}(50)^2 + 9.81(10)) - (0.5\times1.5)((2675\times10^3) + \tfrac{1}{2}(100)^2 + (9.81\times0)) - (0.5\times1.5)((417.46\times10^3) + \tfrac{1}{2}(100)^2 + (9.81\times0))$$

$$0 = \dot{Q}_{cv} - (2.2\times10^6) + (4.74\times10^6) - (2.010375\times10^6) - (0.3168\times10^6)$$

$$= \dot{Q}_{cv} + 0.2128\times10^6 \text{ W}$$

$$\therefore \dot{Q}_{cv} = -0.2128\times10^{+6} \text{ W} = -0.2128 \text{ MW}$$

4 4 **15 / 15**

✓ **- 0 pts** Correct

- 3 pts Q is negative. Heat transfer from the system

- 2 pts Final answer should be in MW

- 5 pts Calculation error. Q = -0.21 MW

- 5 pts Delta PE and Delta KE are different than zero

- 5 pts Wrong unit conversion. Q = -0.21 MW

- 15 pts Missing or wrong answer

86) At $P_1 = 1 bar$ and $x_1 = 1\%$

$$v_1 = v_f(1-x) + v_g(x)$$
$$= (1-0.01)1.0432\times10^{-3} + 0.01\times1.6929$$
$$= 0.018\ m^3/kg$$

$$u_1 = u_f(1-x) + u_g(x)$$
$$= 417.36(1-0.01) + 2506.1(0.01)$$
$$= 438.25\ KJ/kg$$

$$m_1 = \frac{V_1}{v_1} = \frac{0.1\ m^3}{0.018\ m^3/kg} = 5.56\ kg = m_2$$

Stage 1, At 20 bar.

$$v_1 = v_f + x_2(v_g - v_f)$$
$$0.018 = 1.177\times10^{-3} + x_2(0.09963 - 1.177\times10^{-3})$$
$$\therefore x_2 = 0.17$$

$$u_2 = u_g(x) + u_f(1-x)$$
$$= 2600.3(0.17) + 906.44(1-0.17)$$
$$= 1193.93\ KJ/kg.$$

$$Q_1 = m(u_2 - u_1) = 5.56(1193.93 - 438.282)$$
$$= 4201.4\ KJ$$

At stage 2, Saturated vapour at 20 bar.

$$v_3 = v_{g3} = 0.09963\ m^3/kg$$
$$u_3 = 2600.3\ KJ/kg$$
$$h_3 = 2799.5$$

$$m_3 = \frac{V_3}{v_3} = \frac{0.1}{0.09963} = 1.004\ kg$$

$$Q_2 = (m_3u_3 - m_2u_2) - h_3(m_3 - m_2)$$
$$= (1.004\times2600.3 - 5.56\times1193.93) - 2799.5(1.004 - 5.56)$$
$$= 8720.3\ KJ/kg.$$

5 5 15 / 15

✓ **- 0 pts** Correct

- 5 pts Stage 1: Q = mΔu (closed system energy balance) = 4204.2 kJ

- 3 pts Stage 2: correct equation but wrong solution. Q = +8725.5 kJ. Heat transfer to the system.

- 5 pts Stage 2: Q = Δ(m.u) - heΔm = +8725.5 kJ (energy balance one-exit control volume). Heat transfer to the system.

- 15 pts Missing/wrong answer

81) Model

- Ideal gas
- No steady state
- Control volume
- $\dot{Q}_{cv} = \Delta KE = \Delta PE = 0$
- One inlet, two outlet

$\therefore Pv = RT$

$PV = mRT$

$$m_1 = \frac{P_1 V_1}{RT_1} = \frac{100000\ Pa \times 10^5\ m^3}{\frac{8314\ \frac{N\cdot m}{kmol\cdot K}}{28.97\ \frac{kg}{kmol}} \times 290\ K} = 1.2\times10^5\ kg.$$

$$m_2 = \frac{P_2 V_2}{RT_2} = \frac{2100000\ Pa\times10^5\ m^3}{\frac{8314\ \frac{N\cdot m}{kmol\cdot K}}{28.97\ \frac{kg}{kmol}} \times 790\ K} = 9.26\times10^5\ kg.$$

$$\frac{dU_{cv}}{dt} = \overset{0}{\dot{Q}_{cv}} - \dot{W}_{cv} + \dot{m}_i h_i - \overset{0}{\dot{m}_e h_e}$$

$$\frac{dU_{cv}}{dt} = 0 - \dot{W}_{cv} + \dot{m}_i h_i - 0$$

$$\Delta U_{cv} = -\dot{W}_{cv} + \int_1^2 \dot{m}_i h_i$$

$$m_2 u_2 - m_1 u_1 = -\dot{W}_{cv} + h_i (m_2 - m_1)$$

from, table A-22, At: $T = 290K$, $u_1 = 206.91\ \frac{kJ}{kg}$, $h_1 = 290.16\ \frac{kJ}{kg}$.
At $T_2 = 790\ K$, $u_2 = 584.21\ \frac{kJ}{kg}$.

$$\dot{W}_{cv} = m_2(h_1 - u_2) + m_1(u_1 - h_1)$$
$$= 9.26\times10^5(290.16 - 584.21) + 1.2\times10^5(206.91 - 290.16)$$
$$= -282280300\ J \times \frac{1}{10^6}$$
$$= -282.28\ GJ$$

6 6 **17.5 / 20**

- **- 0 pts** Correct
- **- 1.5 pts** Wrong unit conversion. m1 = 1.2 x 10^5 kg and m2 = 9.3 x 10^5 kg
- **- 5 pts** Wrong energy balance. Should use Eq 4.28
- **- 5 pts** W = 282 GJ
- ✓ **- 2.5 pts** **Work is positive (done by the compressor)**
- **- 20 pts** Wrong answer
- **- 0 pts** Click here to replace this description.